

Section 13.1: Vector Functions and Space Curves

Vector Functions

MTH 310
Calculus III

Vector Functions

In Calc I/II, functions take a number in and give a number back: $f(t) = t^2$.

Now we let the output be a **vector** instead of a scalar.

Definition 1: Vector Function

A **vector function** (or **vector-valued function**) is a function whose input is a scalar t and whose output is a vector:

$$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

The functions f, g, h are called the **component functions**.

As t varies, the tip of $\vec{r}(t)$ traces out a **space curve** C .

[Animated Space Curve](https://www.desmos.com/3d/2afd96drie) [https://www.desmos.com/3d/2afd96drie] — An animated figure showing the space curve $\vec{r}(t) = \langle \sin t, \cos t, \sin 2t \rangle$ with a position vector from the origin to the moving point.

A vector function is just a way of packaging **parametric equations** into a single object.

Why Vector Functions?

Think of t as **time**. Then $\vec{r}(t)$ gives the **position** of a moving object at time t .

- The path it traces is the **trajectory** (a space curve).

Applications everywhere:

- Roller coasters, planetary orbits, particle physics
- Robot arm paths, drone flight plans
- Blood flow through arteries

Chapter 13 is about **calculus on curves** — the same derivative and integral ideas from Calc I, but now applied to vector functions.

Example 1: Sketch $\vec{r}(t) = \langle t^3, t^2 \rangle$

Strategy for sketching a vector function:

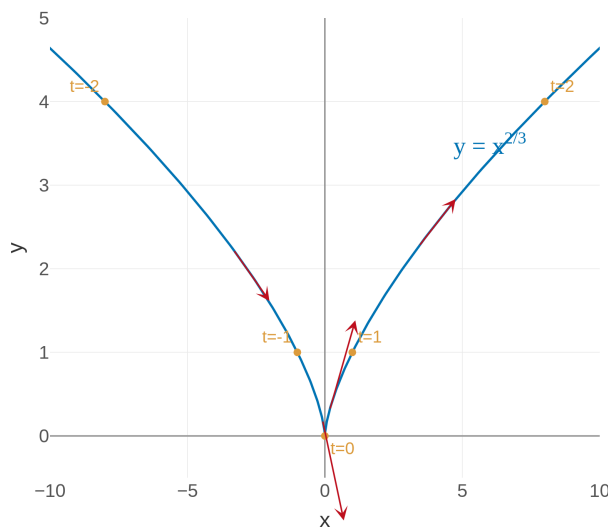
1. Make a table of values
2. Try to eliminate t to recognize the Cartesian curve
3. Mark the direction of travel (increasing t)

Step 1 — Make a table. Plug in a few values of t :

t	$x = t^3$	$y = t^2$
-2	-8	4
-1	-1	1
0	0	0
1	1	1
2	8	4

Step 2 — Eliminate the parameter. From $x = t^3$ we get $t = x^{1/3}$, so $y = (x^{1/3})^2 = x^{2/3}$.

Step 3 — Direction of travel. As t goes from -2 to 2 , the point moves left-to-right through the origin.



Desmos 3D [<https://www.desmos.com/3d>]

The curve $y = x^{2/3}$ has a **cusp** at the origin. The parametrization tells us *which way* to travel along the curve.

Your Turn: Sketch $\vec{r}(t) = \langle 2 \cos t, 2 \sin t \rangle$

Use the same strategy: make a table, eliminate t , identify the curve.

Try it before looking at the solution.

Space Curves in 3D

In 3D, a vector function $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ traces out a **space curve**.

We can still make tables, but now we plot in xyz -space.

Common strategy: Look at what the components tell you.

- Do any components involve trig? That hints at circular motion.
- Is one component linear (like $y = t$)? That means steady motion in one direction.
- Can you combine components to recognize a surface (like $x^2 + z^2 = 1$)?

The component functions are the clues — they tell you what surface the curve lives on and how it moves.

Example 2: Sketch $\vec{r}(t) = \langle \sin \pi t, t, \cos \pi t \rangle$

Step 1 — Identify the cylinder. Look at which components involve trig:

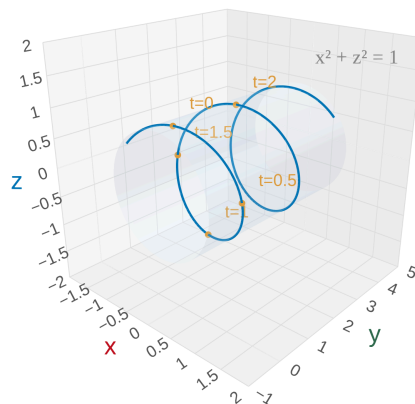
$$x^2 + z^2 = \sin^2(\pi t) + \cos^2(\pi t) = 1$$

The curve lives on the cylinder $x^2 + z^2 = 1$ (radius 1, centered on the y -axis).

Step 2 — What does y do? $y = t$ increases steadily $\rightarrow$ the curve spirals **upward** along the y -axis.

Step 3 — How fast does it wind? The argument πt completes one full cycle when πt goes from 0 to 2π , i.e., when t goes from 0 to 2. So **one full turn every 2 units** along the y -axis.

t	$x = \sin \pi t$	$y = t$	$z = \cos \pi t$
0	0	0	1
$\frac{1}{2}$	1	$\frac{1}{2}$	0
1	0	1	-1
$\frac{3}{2}$	-1	$\frac{3}{2}$	0
2	0	2	1



Desmos 3D [<https://www.desmos.com/3d>]

This is a **helix** — trig + trig + linear = spiral. Think of wrapping a string around a paper towel tube.

Where Does a Curve Meet a Surface?

A common question: **where does a space curve intersect a surface?**

The strategy is straightforward:

1. **Substitute** the components $x = f(t)$, $y = g(t)$, $z = h(t)$ into the surface equation.
2. **Solve** the resulting equation for t .
3. **Plug back** into $\vec{r}(t)$ to get the point(s).

This reduces a 3D geometry problem to **solving an equation in one variable t** .

Example 3: Where does $\vec{r}(t) = \langle \sin t, \cos t, t \rangle$ meet $x^2 + y^2 + z^2 = 5$?

Step 1 — Substitute the components into the sphere equation:

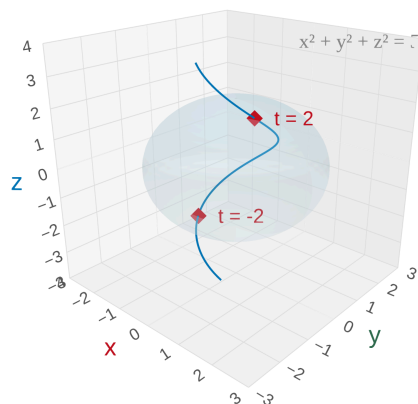
$$\sin^2 t + \cos^2 t + t^2 = 5$$

Step 2 — Simplify. The Pythagorean identity $\sin^2 t + \cos^2 t = 1$ collapses two terms:

$$1 + t^2 = 5 \quad \Rightarrow \quad t^2 = 4 \quad \Rightarrow \quad t = \pm 2$$

Step 3 — Find the points. Plug $t = 2$ and $t = -2$ back into $\vec{r}(t)$:

- $\vec{r}(2) = \langle \sin 2, \cos 2, 2 \rangle \approx \langle 0.909, -0.416, 2 \rangle$
- $\vec{r}(-2) = \langle \sin(-2), \cos(-2), -2 \rangle \approx \langle -0.909, -0.416, -2 \rangle$



Desmos 3D [<https://www.desmos.com/3d>]

The helix pierces the sphere at exactly **two points**, at $t = \pm 2$. The Pythagorean identity did the heavy lifting!

Your Turn: Where does $\vec{r}(t) = \langle t, 2t, 3t \rangle$ meet $x^2 + y^2 + z^2 = 14$?

Use the same three steps: substitute, solve for t , plug back in.

Try it before looking at the solution.

Homework

Section 13.1: 3, 5, 7, 11, 15, 19, 25–30